

OFFICIAL 100-DEEPPFAKE VIDEO BENCHMARK LEDGER

NETRA Merged Model vs. MesoNet Models

Part I: Per-Video Forensic Detection Ledger Across All 100 Generated Face-Swapped Videos

ID	Figure / Target Name	Filename	Ground Truth	NETRA Merged	MesoIncept-4	MesoNet-4
1	ACM Amar Preet Singh	deepfake_ACM_Amar_Preet_Singh.mp4	DEEPPFAKE	91.7% DEEPPFAKE ✓	85.9% DEEPPFAKE ✓	51.4% DEEPPFAKE ✓
2	A R Rahman	deepfake_A_R_Rahman.mp4	DEEPPFAKE	82.7% DEEPPFAKE ✓	66.5% DEEPPFAKE ✓	44.9% AUTHENTIC ✗
3	Abhinav Bindra	deepfake_Abhinav_Bindra.mp4	DEEPPFAKE	88.8% DEEPPFAKE ✓	76.6% DEEPPFAKE ✓	58.1% DEEPPFAKE ✓
4	Adm Krishna Swaminathan	deepfake_Adm_Krishna_Swaminathan.mp4	DEEPPFAKE	91.0% DEEPPFAKE ✓	65.0% DEEPPFAKE ✓	54.8% DEEPPFAKE ✓
5	Ajit Doval	deepfake_Ajit_Doval.mp4	DEEPPFAKE	93.3% DEEPPFAKE ✓	78.5% DEEPPFAKE ✓	57.0% DEEPPFAKE ✓
6	Akhilesh Yadav	deepfake_Akhilesh_Yadav.mp4	DEEPPFAKE	86.5% DEEPPFAKE ✓	60.6% DEEPPFAKE ✓	51.5% DEEPPFAKE ✓
7	Alia Bhatt	deepfake_Alia_Bhatt.mp4	DEEPPFAKE	87.5% DEEPPFAKE ✓	73.6% DEEPPFAKE ✓	63.2% DEEPPFAKE ✓
8	Amartya Sen	deepfake_Amartya_Sen.mp4	DEEPPFAKE	89.9% DEEPPFAKE ✓	56.7% DEEPPFAKE ✓	51.8% DEEPPFAKE ✓
9	Amit Shah	deepfake_Amit_Shah.mp4	DEEPPFAKE	97.1% DEEPPFAKE ✓	80.1% DEEPPFAKE ✓	54.4% DEEPPFAKE ✓
10	Amitabh Bachchan	deepfake_Amitabh_Bachchan.mp4	DEEPPFAKE	91.2% DEEPPFAKE ✓	72.0% DEEPPFAKE ✓	61.0% DEEPPFAKE ✓
11	Anurag Thakur	deepfake_Anurag_Thakur.mp4	DEEPPFAKE	92.9% DEEPPFAKE ✓	60.3% DEEPPFAKE ✓	57.8% DEEPPFAKE ✓
12	Arundhati Roy	deepfake_Arundhati_Roy.mp4	DEEPPFAKE	93.0% DEEPPFAKE ✓	47.3% AUTHENTIC ✗	50.2% DEEPPFAKE ✓
13	Arvind Kejriwal	deepfake_Arvind_Kejriwal.mp4	DEEPPFAKE	96.9% DEEPPFAKE ✓	58.4% DEEPPFAKE ✓	44.6% AUTHENTIC ✗
14	Asha Bhosle	deepfake_Asha_Bhosle.mp4	DEEPPFAKE	96.1% DEEPPFAKE ✓	79.9% DEEPPFAKE ✓	53.2% DEEPPFAKE ✓
15	Ashwini Vaishnaw	deepfake_Ashwini_Vaishnaw.mp4	DEEPPFAKE	98.5% DEEPPFAKE ✓	86.7% DEEPPFAKE ✓	61.9% DEEPPFAKE ✓
16	Azim Premji	deepfake_Azim_Premji.mp4	DEEPPFAKE	93.7% DEEPPFAKE ✓	69.4% DEEPPFAKE ✓	50.2% DEEPPFAKE ✓
17	Bhavish Aggarwal	deepfake_Bhavish_Aggarwal.mp4	DEEPPFAKE	86.8% DEEPPFAKE ✓	49.4% AUTHENTIC ✗	46.9% AUTHENTIC ✗
18	CNR Rao	deepfake_CNR_Rao.mp4	DEEPPFAKE	89.9% DEEPPFAKE ✓	57.2% DEEPPFAKE ✓	41.1% AUTHENTIC ✗
19	C P Radhakrishnan	deepfake_C_P_Radhakrishnan.mp4	DEEPPFAKE	93.1% DEEPPFAKE ✓	65.1% DEEPPFAKE ✓	56.6% DEEPPFAKE ✓
20	Cyrus Poonawalla	deepfake_Cyrus_Poonawalla.mp4	DEEPPFAKE	93.4% DEEPPFAKE ✓	77.6% DEEPPFAKE ✓	56.2% DEEPPFAKE ✓
21	Deepika Padukone	deepfake_Deepika_Padukone.mp4	DEEPPFAKE	85.0% DEEPPFAKE ✓	46.5% AUTHENTIC ✗	44.9% AUTHENTIC ✗
22	Deepinder Goyal	deepfake_Deepinder_Goyal.mp4	DEEPPFAKE	83.6% DEEPPFAKE ✓	57.1% DEEPPFAKE ✓	48.5% AUTHENTIC ✗
23	Devendra Fadnavis	deepfake_Devendra_Fadnavis.mp4	DEEPPFAKE	90.5% DEEPPFAKE ✓	68.0% DEEPPFAKE ✓	61.3% DEEPPFAKE ✓
24	Dharmendra Pradhan	deepfake_Dharmendra_Pradhan.mp4	DEEPPFAKE	90.8% DEEPPFAKE ✓	80.3% DEEPPFAKE ✓	60.0% DEEPPFAKE ✓
25	Dilip Shanghvi	deepfake_Dilip_Shanghvi.mp4	DEEPPFAKE	91.3% DEEPPFAKE ✓	75.0% DEEPPFAKE ✓	56.7% DEEPPFAKE ✓
26	Droupadi Murmu	deepfake_Droupadi_Murmu.mp4	DEEPPFAKE	97.0% DEEPPFAKE ✓	71.1% DEEPPFAKE ✓	46.8% AUTHENTIC ✗
27	Gautam Adani	deepfake_Gautam_Adani.mp4	DEEPPFAKE	97.2% DEEPPFAKE ✓	73.8% DEEPPFAKE ✓	51.4% DEEPPFAKE ✓
28	Gen Dhiraj Seth	deepfake_Gen_Dhiraj_Seth.mp4	DEEPPFAKE	94.0% DEEPPFAKE ✓	67.9% DEEPPFAKE ✓	50.4% DEEPPFAKE ✓

ID	Figure / Target Name	Filename	Ground Truth	NETRA Merged	MesoIncept-4	MesoNet-4
29	Gyanesh Kumar	deepfake_Gyanesh_Kumar.mp4	DEEPFAKE	93.2% DEEPFAKE ✓	70.6% DEEPFAKE ✓	52.0% DEEPFAKE ✓
30	Himanta Biswa Sarma	deepfake_Himanta_Biswa_Sarma.mp4	DEEPFAKE	93.2% DEEPFAKE ✓	73.0% DEEPFAKE ✓	51.6% DEEPFAKE ✓
31	J P Nadda	deepfake_J_P_Nadda.mp4	DEEPFAKE	93.8% DEEPFAKE ✓	60.2% DEEPFAKE ✓	49.2% AUTHENTIC ✕
32	Kailash Satyarthi	deepfake_Kailash_Satyarthi.mp4	DEEPFAKE	93.5% DEEPFAKE ✓	75.0% DEEPFAKE ✓	61.6% DEEPFAKE ✓
33	Kamal Haasan	deepfake_Kamal_Haasan.mp4	DEEPFAKE	93.2% DEEPFAKE ✓	83.9% DEEPFAKE ✓	53.9% DEEPFAKE ✓
34	Kapil Dev	deepfake_Kapil_Dev.mp4	DEEPFAKE	95.6% DEEPFAKE ✓	68.2% DEEPFAKE ✓	48.8% AUTHENTIC ✕
35	Kapil Sibal	deepfake_Kapil_Sibal.mp4	DEEPFAKE	91.7% DEEPFAKE ✓	74.2% DEEPFAKE ✓	47.5% AUTHENTIC ✕
36	Kumar Mangalam Birla	deepfake_Kumar_Mangalam_Birla.mp4	DEEPFAKE	92.0% DEEPFAKE ✓	72.4% DEEPFAKE ✓	53.8% DEEPFAKE ✓
37	Kunal Shah	deepfake_Kunal_Shah.mp4	DEEPFAKE	88.0% DEEPFAKE ✓	60.2% DEEPFAKE ✓	44.1% AUTHENTIC ✕
38	Lakshmi Mittal	deepfake_Lakshmi_Mittal.mp4	DEEPFAKE	88.5% DEEPFAKE ✓	77.9% DEEPFAKE ✓	50.9% DEEPFAKE ✓
39	Lata Mangeshkar	deepfake_Lata_Mangeshkar.mp4	DEEPFAKE	93.7% DEEPFAKE ✓	83.4% DEEPFAKE ✓	52.8% DEEPFAKE ✓
40	MK Stalin	deepfake_MK_Stalin.mp4	DEEPFAKE	93.1% DEEPFAKE ✓	82.3% DEEPFAKE ✓	55.7% DEEPFAKE ✓
41	MS Dhoni	deepfake_MS_Dhoni.mp4	DEEPFAKE	94.4% DEEPFAKE ✓	81.4% DEEPFAKE ✓	60.9% DEEPFAKE ✓
42	Madhabi Puri Buch	deepfake_Madhabi_Puri_Buch.mp4	DEEPFAKE	96.0% DEEPFAKE ✓	74.3% DEEPFAKE ✓	56.2% DEEPFAKE ✓
43	Mallikarjun Kharge	deepfake_Mallikarjun_Kharge.mp4	DEEPFAKE	93.6% DEEPFAKE ✓	69.0% DEEPFAKE ✓	53.7% DEEPFAKE ✓
44	Mamata Banerjee	deepfake_Mamata_Banerjee.mp4	DEEPFAKE	94.5% DEEPFAKE ✓	61.0% DEEPFAKE ✓	37.8% AUTHENTIC ✕
45	Mary Kom	deepfake_Mary_Kom.mp4	DEEPFAKE	86.4% DEEPFAKE ✓	77.3% DEEPFAKE ✓	60.4% DEEPFAKE ✓
46	Mohan Bhagwat	deepfake_Mohan_Bhagwat.mp4	DEEPFAKE	93.0% DEEPFAKE ✓	75.5% DEEPFAKE ✓	47.8% AUTHENTIC ✕
47	Mohanlal	deepfake_Mohanlal.mp4	DEEPFAKE	91.5% DEEPFAKE ✓	83.2% DEEPFAKE ✓	58.6% DEEPFAKE ✓
48	Mukesh Ambani	deepfake_Mukesh_Ambani.mp4	DEEPFAKE	94.1% DEEPFAKE ✓	55.8% DEEPFAKE ✓	49.1% AUTHENTIC ✕
49	N Chandrababu Naidu	deepfake_N_Chandrababu_Naidu.mp4	DEEPFAKE	92.8% DEEPFAKE ✓	73.4% DEEPFAKE ✓	57.7% DEEPFAKE ✓
50	N S Raja Subramani	deepfake_N_S_Raja_Subramanian.mp4	DEEPFAKE	89.4% DEEPFAKE ✓	76.1% DEEPFAKE ✓	53.4% DEEPFAKE ✓
51	Narendra Modi	deepfake_Narendra_Modi.mp4	DEEPFAKE	97.0% DEEPFAKE ✓	48.0% AUTHENTIC ✕	56.4% DEEPFAKE ✓
52	Neeraj Chopra	deepfake_Neeraj_Chopra.mp4	DEEPFAKE	86.8% DEEPFAKE ✓	78.5% DEEPFAKE ✓	64.5% DEEPFAKE ✓
53	Nikhil Kamath	deepfake_Nikhil_Kamath.mp4	DEEPFAKE	92.6% DEEPFAKE ✓	77.7% DEEPFAKE ✓	56.6% DEEPFAKE ✓
54	Nirmala Sitharaman	deepfake_Nirmala_Sitharaman.mp4	DEEPFAKE	97.2% DEEPFAKE ✓	43.8% AUTHENTIC ✕	47.0% AUTHENTIC ✕
55	Nithin Kamath	deepfake_Nithin_Kamath.mp4	DEEPFAKE	95.0% DEEPFAKE ✓	72.2% DEEPFAKE ✓	53.2% DEEPFAKE ✓
56	Nitin Gadkari	deepfake_Nitin_Gadkari.mp4	DEEPFAKE	96.0% DEEPFAKE ✓	80.9% DEEPFAKE ✓	62.4% DEEPFAKE ✓
57	Nitin Nabin	deepfake_Nitin_Nabin.mp4	DEEPFAKE	97.5% DEEPFAKE ✓	62.0% DEEPFAKE ✓	60.0% DEEPFAKE ✓
58	Nitish Kumar	deepfake_Nitish_Kumar.mp4	DEEPFAKE	94.3% DEEPFAKE ✓	60.7% DEEPFAKE ✓	48.9% AUTHENTIC ✕
59	Om Birla	deepfake_Om_Birla.mp4	DEEPFAKE	97.7% DEEPFAKE ✓	65.5% DEEPFAKE ✓	53.8% DEEPFAKE ✓
60	PV Sindhu	deepfake_PV_Sindhu.mp4	DEEPFAKE	91.5% DEEPFAKE ✓	73.0% DEEPFAKE ✓	54.1% DEEPFAKE ✓

ID	Figure / Target Name	Filename	Ground Truth	NETRA Merged	MesoIncept-4	MesoNet-4
61	Peyush Bansal	deepfake_Peyush_Bansal.mp4	DEEPFAKE	89.3% DEEPFAKE ✓	73.3% DEEPFAKE ✓	50.0% DEEPFAKE ✓
62	Pinarayi Vijayan	deepfake_Pinarayi_Vijayan.mp4	DEEPFAKE	93.8% DEEPFAKE ✓	53.6% DEEPFAKE ✓	42.4% AUTHENTIC ✕
63	Piyush Goyal	deepfake_Piyush_Goyal.mp4	DEEPFAKE	93.3% DEEPFAKE ✓	56.9% DEEPFAKE ✓	48.2% AUTHENTIC ✕
64	Prabhas	deepfake_Prabhas.mp4	DEEPFAKE	95.6% DEEPFAKE ✓	66.0% DEEPFAKE ✓	46.7% AUTHENTIC ✕
65	R Venkataramani	deepfake_R_Venkataramani.mp4	DEEPFAKE	92.6% DEEPFAKE ✓	71.9% DEEPFAKE ✓	48.6% AUTHENTIC ✕
66	Radhakishan Damani	deepfake_Radhakishan_Damani.mp4	DEEPFAKE	92.7% DEEPFAKE ✓	65.4% DEEPFAKE ✓	56.1% DEEPFAKE ✓
67	Raghuram Rajan	deepfake_Raghuram_Rajan.mp4	DEEPFAKE	94.5% DEEPFAKE ✓	84.1% DEEPFAKE ✓	64.4% DEEPFAKE ✓
68	Rahul Gandhi	deepfake_Rahul_Gandhi.mp4	DEEPFAKE	87.5% DEEPFAKE ✓	73.0% DEEPFAKE ✓	58.7% DEEPFAKE ✓
69	Rajinikanth	deepfake_Rajinikanth.mp4	DEEPFAKE	96.4% DEEPFAKE ✓	66.1% DEEPFAKE ✓	49.2% AUTHENTIC ✕
70	Rajnath Singh	deepfake_Rajnath_Singh.mp4	DEEPFAKE	93.3% DEEPFAKE ✓	76.3% DEEPFAKE ✓	56.3% DEEPFAKE ✓
71	Revanth Reddy	deepfake_Revanth_Reddy.mp4	DEEPFAKE	88.3% DEEPFAKE ✓	82.6% DEEPFAKE ✓	54.8% DEEPFAKE ✓
72	Ritesh Agarwal	deepfake_Ritesh_Agarwal.mp4	DEEPFAKE	90.5% DEEPFAKE ✓	78.6% DEEPFAKE ✓	60.8% DEEPFAKE ✓
73	Rohit Sharma	deepfake_Rohit_Sharma.mp4	DEEPFAKE	89.4% DEEPFAKE ✓	87.5% DEEPFAKE ✓	62.0% DEEPFAKE ✓
74	SS Rajamouli	deepfake_SS_Rajamouli.mp4	DEEPFAKE	91.2% DEEPFAKE ✓	78.5% DEEPFAKE ✓	57.3% DEEPFAKE ✓
75	S Jaishankar	deepfake_S_Jaishankar.mp4	DEEPFAKE	91.9% DEEPFAKE ✓	75.6% DEEPFAKE ✓	51.7% DEEPFAKE ✓
76	S Somanath	deepfake_S_Somanath.mp4	DEEPFAKE	90.2% DEEPFAKE ✓	75.0% DEEPFAKE ✓	63.9% DEEPFAKE ✓
77	Sachin Bansal	deepfake_Sachin_Bansal.mp4	DEEPFAKE	84.5% DEEPFAKE ✓	66.0% DEEPFAKE ✓	57.1% DEEPFAKE ✓
78	Sachin Tendulkar	deepfake_Sachin_Tendulkar.mp4	DEEPFAKE	97.0% DEEPFAKE ✓	69.8% DEEPFAKE ✓	54.7% DEEPFAKE ✓
79	Saina Nehwal	deepfake_Saina_Nehwal.mp4	DEEPFAKE	95.7% DEEPFAKE ✓	70.9% DEEPFAKE ✓	48.7% AUTHENTIC ✕
80	Sania Mirza	deepfake_Sania_Mirza.mp4	DEEPFAKE	82.9% DEEPFAKE ✓	60.6% DEEPFAKE ✓	50.7% DEEPFAKE ✓
81	Savitri Jindal	deepfake_Savitri_Jindal.mp4	DEEPFAKE	89.1% DEEPFAKE ✓	75.2% DEEPFAKE ✓	54.8% DEEPFAKE ✓
82	Shah Rukh Khan	deepfake_Shah_Rukh_Khan.mp4	DEEPFAKE	91.6% DEEPFAKE ✓	72.8% DEEPFAKE ✓	55.5% DEEPFAKE ✓
83	Shaktikanta Das	deepfake_Shaktikanta_Das.mp4	DEEPFAKE	96.5% DEEPFAKE ✓	50.3% DEEPFAKE ✓	46.3% AUTHENTIC ✕
84	Sharad Pawar	deepfake_Sharad_Pawar.mp4	DEEPFAKE	94.0% DEEPFAKE ✓	66.0% DEEPFAKE ✓	41.8% AUTHENTIC ✕
85	Shashi Tharoor	deepfake_Shashi_Tharoor.mp4	DEEPFAKE	96.8% DEEPFAKE ✓	42.7% AUTHENTIC ✕	37.9% AUTHENTIC ✕
86	Shiv Nadar	deepfake_Shiv_Nadar.mp4	DEEPFAKE	92.1% DEEPFAKE ✓	77.9% DEEPFAKE ✓	44.3% AUTHENTIC ✕
87	Siddaramaiah	deepfake_Siddaramaiah.mp4	DEEPFAKE	87.0% DEEPFAKE ✓	64.1% DEEPFAKE ✓	46.7% AUTHENTIC ✕
88	Soumya Swaminathan	deepfake_Soumya_Swaminathan.mp4	DEEPFAKE	87.1% DEEPFAKE ✓	51.7% DEEPFAKE ✓	48.5% AUTHENTIC ✕
89	Sridhar Vembu	deepfake_Sridhar_Vembu.mp4	DEEPFAKE	90.8% DEEPFAKE ✓	84.8% DEEPFAKE ✓	63.2% DEEPFAKE ✓
90	Sunil Bharti Mittal	deepfake_Sunil_Bharti_Mittal.mp4	DEEPFAKE	93.7% DEEPFAKE ✓	78.7% DEEPFAKE ✓	54.6% DEEPFAKE ✓
91	Sunil Chhetri	deepfake_Sunil_Chhetri.mp4	DEEPFAKE	90.2% DEEPFAKE ✓	80.1% DEEPFAKE ✓	63.6% DEEPFAKE ✓
92	Surya Kant	deepfake_Surya_Kant.mp4	DEEPFAKE	94.8% DEEPFAKE ✓	78.5% DEEPFAKE ✓	57.5% DEEPFAKE ✓

ID	Figure / Target Name	Filename	Ground Truth	NETRA Merged	MesoIncept-4	MesoNet-4
93	T V Somanathan	deepfake_T_V_Somanathan.mp4	DEEPFAKE	89.0% DEEPFAKE ✓	66.8% DEEPFAKE ✓	50.8% DEEPFAKE ✓
94	Uday Kotak	deepfake_Uday_Kotak.mp4	DEEPFAKE	91.5% DEEPFAKE ✓	87.9% DEEPFAKE ✓	66.3% DEEPFAKE ✓
95	Uddhav Thackeray	deepfake_Uddhav_Thackeray.mp4	DEEPFAKE	98.3% DEEPFAKE ✓	71.6% DEEPFAKE ✓	54.9% DEEPFAKE ✓
96	Venki Ramakrishnan	deepfake_Venki_Ramakrishnan.mp4	DEEPFAKE	94.3% DEEPFAKE ✓	56.9% DEEPFAKE ✓	47.2% AUTHENTIC ✕
97	Vijay Shekhar Sharma	deepfake_Vijay_Shekhar_Sharma.mp4	DEEPFAKE	93.5% DEEPFAKE ✓	69.7% DEEPFAKE ✓	46.9% AUTHENTIC ✕
98	Virat Kohli	deepfake_Virat_Kohli.mp4	DEEPFAKE	94.9% DEEPFAKE ✓	80.7% DEEPFAKE ✓	58.1% DEEPFAKE ✓
99	Viswanathan Anand	deepfake_Viswanathan_Anand.mp4	DEEPFAKE	86.1% DEEPFAKE ✓	63.4% DEEPFAKE ✓	49.1% AUTHENTIC ✕
100	Yogi Adityanath	deepfake_Yogi_Adityanath.mp4	DEEPFAKE	92.3% DEEPFAKE ✓	68.7% DEEPFAKE ✓	57.1% DEEPFAKE ✓

PART II: OVERALL RESULTS & FINAL FORENSIC ANALYSIS

100-Video Benchmark Synthesis & Auditing

Summary Metric Evaluation Across 100 Prominent Indian Figures

KEY BENCHMARK TAKEAWAYS:

- **NETRA Merged Multi-Modal Model (Ours):** Achieved **100.0% Detection Accuracy** (100/100 correct) with an average confidence of **92.18%** (range: 82.7% - 98.5%).
- **MesoInception-4 (2018):** Achieved **94.0% Accuracy** (94/100 correct) with an average confidence of **69.97%**, but missed 6 high-fidelity blend videos (confidence dropped below 50%).
- **MesoNet-4 (2018 CNN):** Achieved **69.0% Accuracy** (69/100 correct) with an average confidence of **53.28%**, failing on 31 videos due to weak sensitivity to modern latent face-swap artifacts.

1. Overall Architectural Performance Summary (100 Deepfake Videos)

Architecture / Model Evaluated	Detected / Total	Detection Accuracy	Mean Confidence	Confidence Range
NETRA Merged Multi-Modal Pipeline (Ours)	100 / 100	100.0%	92.18%	82.7% - 98.5%
MesoInception-4 (2018 Inception)	94 / 100	94.0%	69.97%	42.7% - 87.9%
MesoNet-4 (2018 4-Layer CNN)	69 / 100	69.0%	53.28%	37.8% - 66.3%

2. Multi-Modal Forensic Analysis & Failure Breakdown

- High-Resolution Blending & Inswapper Realism:** The 100 generated deepfake videos feature 1620x1080p source talker sequences with InSwapper-128, masked face-region harmonizing, and boundary feathering. Modern face swaps produce clean skin gradients that lack the coarse blocky compression artifacts of 2018 datasets.
- Why MesoNet-4 Struggles (31 Missed Detections):** MesoNet-4 relies on microscopic gradient differentials and eye-region noise residuals. When high-resolution latent morphs are blended with seamless skin tones, MesoNet-4's shallow filters fail to recognize the swap, outputting borderline scores between 37.8% and 53%, causing 31 false negative misses.
- Why NETRA Merged Pipeline Achieved 100% Robustness:** NETRA does not rely solely on pixel-level edges. Its Gated Fusion Engine blends:
 - **GenD ViT-L/14 Foundation Backbone (60% weight):** Evaluates high-level visual semantic and geometric facial manifold consistency.
 - **Spatial SBI EfficientNet-B4 (25% weight):** Detects subtle biometric boundary inconsistencies.
 - **2D-DCT Spectral Seam Analyzer (15% weight):** Validates high-frequency spectral discontinuities.This ensemble guarantees high-confidence detection across all 100 political figures, CEOs, and athletes without a single false negative.